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(54) **REMOVABLE TOOTHBRUSH HEAD FOR  
ELECTRIC TOOTHBRUSH HANDLE**

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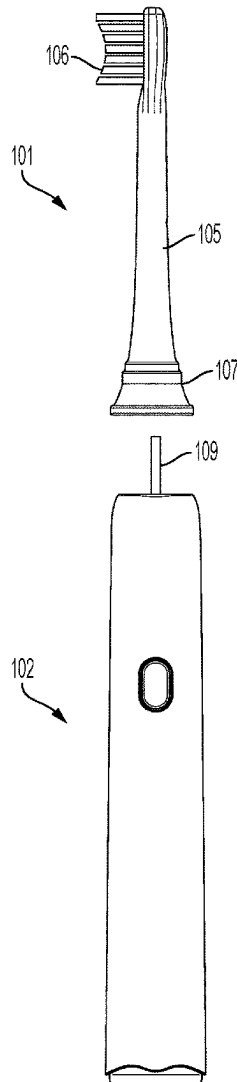
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(57)

**ABSTRACT**

A toothbrush head includes a bristle tray, a base having an opening for receiving an electric toothbrush a drive shaft, a stem that connects the bristle tray to the base, and a coupling member. The coupling member does not require a spring to contact the drive shaft or stem. In some embodiments, the coupling member includes an interior sidewall that surrounds a chamber for receiving the drive shaft and a receptacle positioned to align with an opening of the stem, and a pin is positioned within the opening and the receptacle. In other embodiments, the coupling member includes a relatively narrower section positioned to align with the stem and a relatively wider section positioned to align with the base, and when the drive shaft is inserted into the coupling member, the drive shaft will extend from the relatively narrower section and into an interior cavity within the stem.



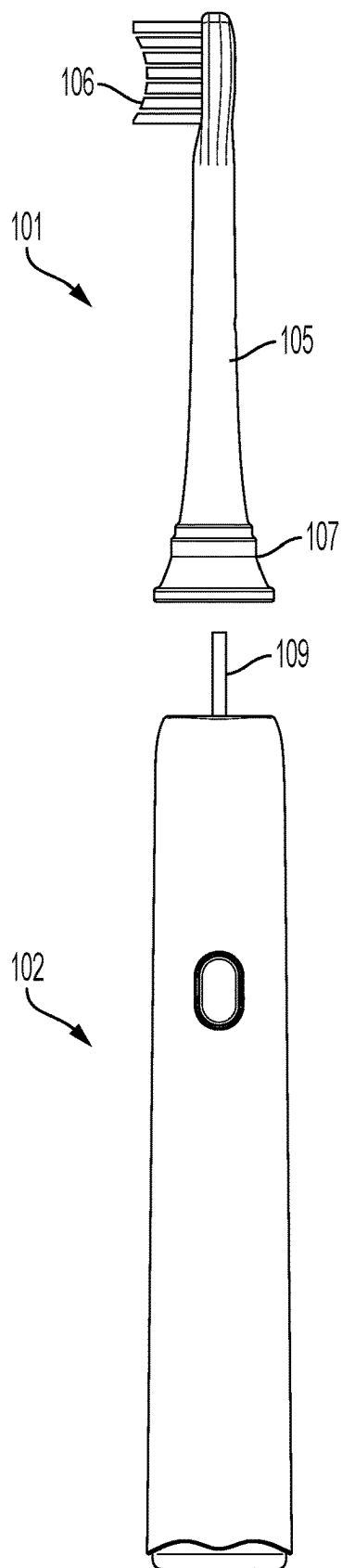


FIG. 1

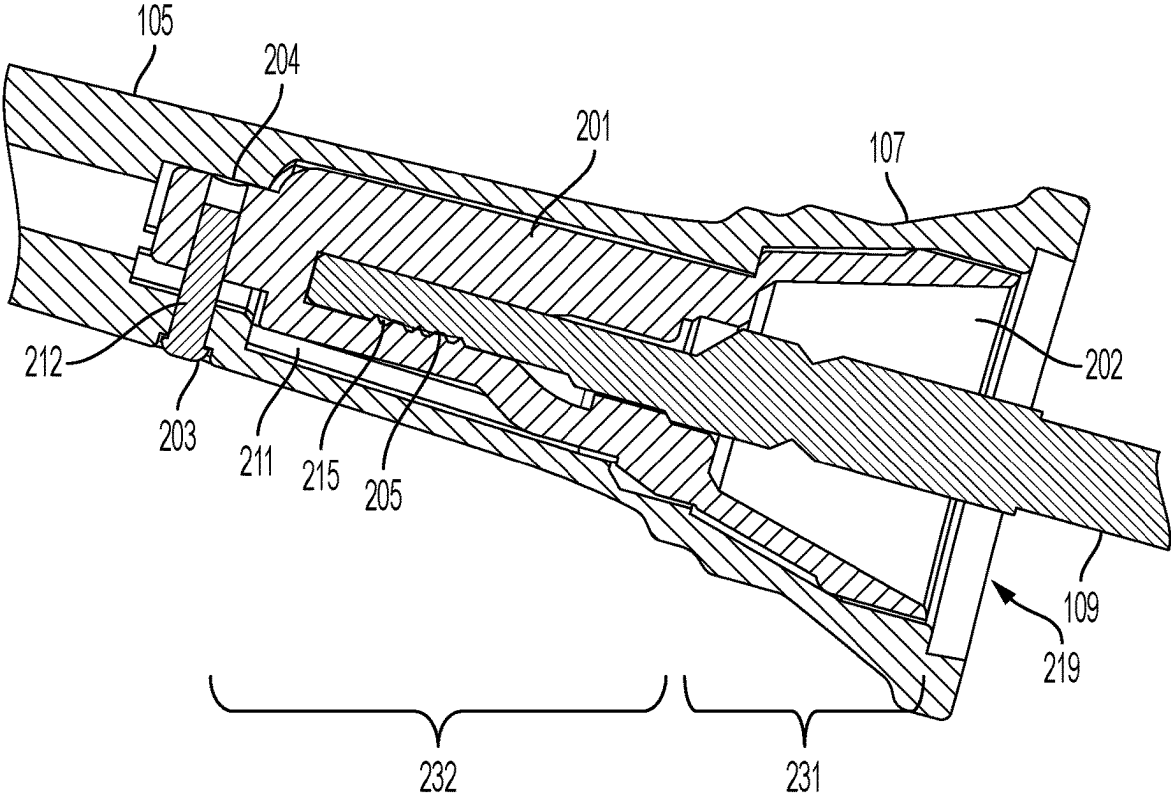
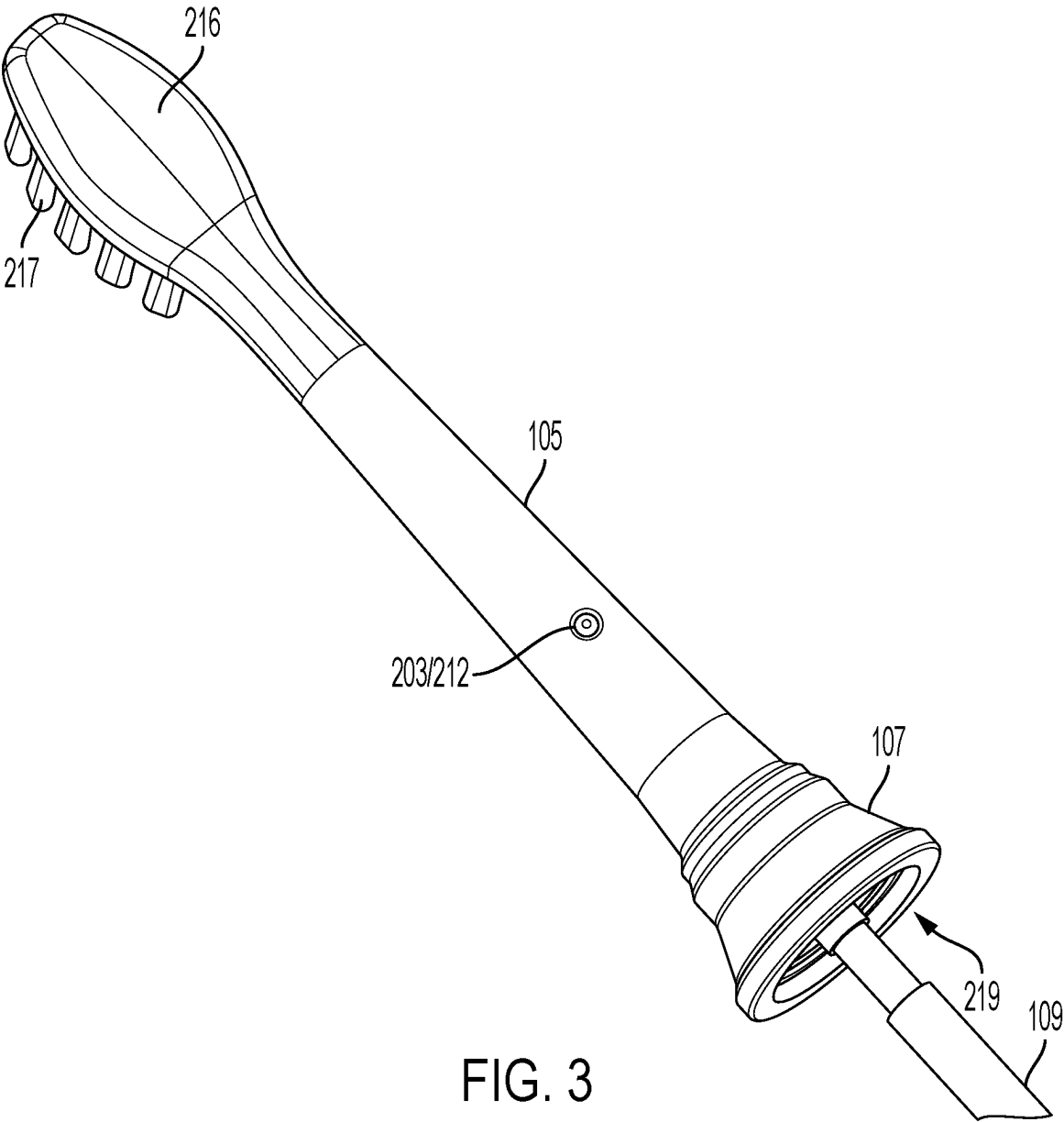


FIG. 2



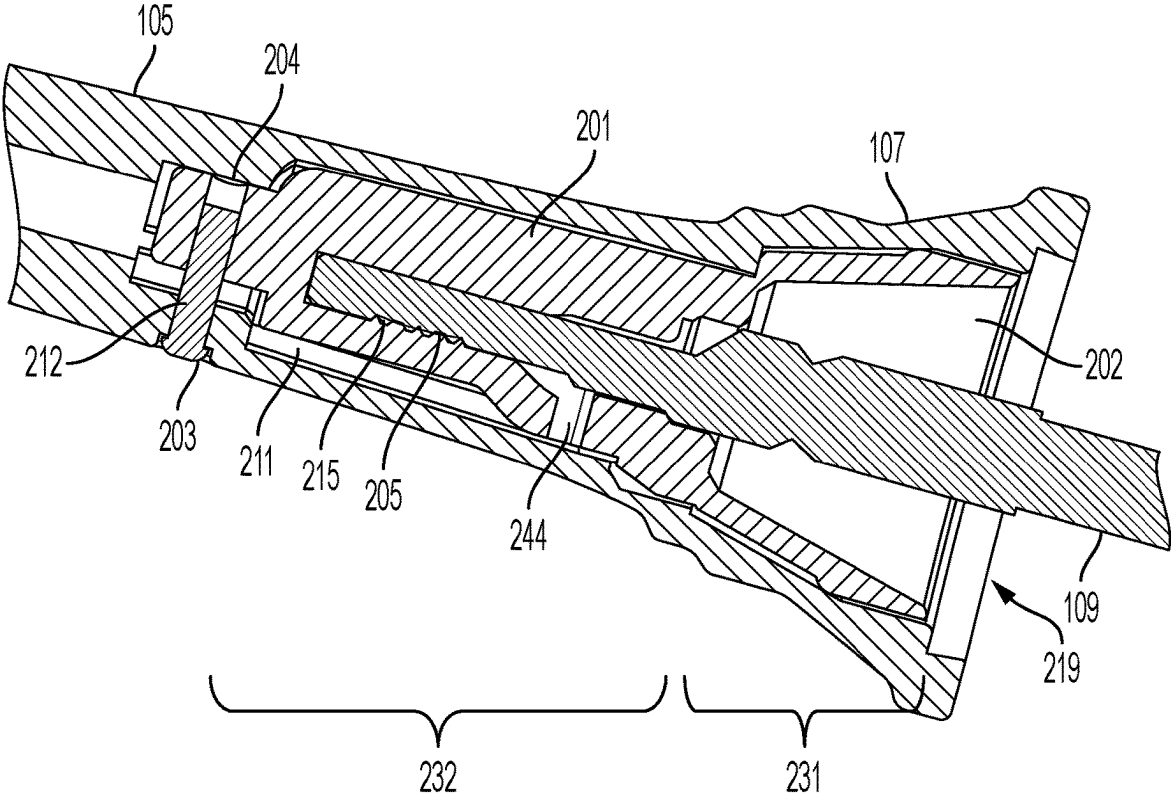


FIG. 4

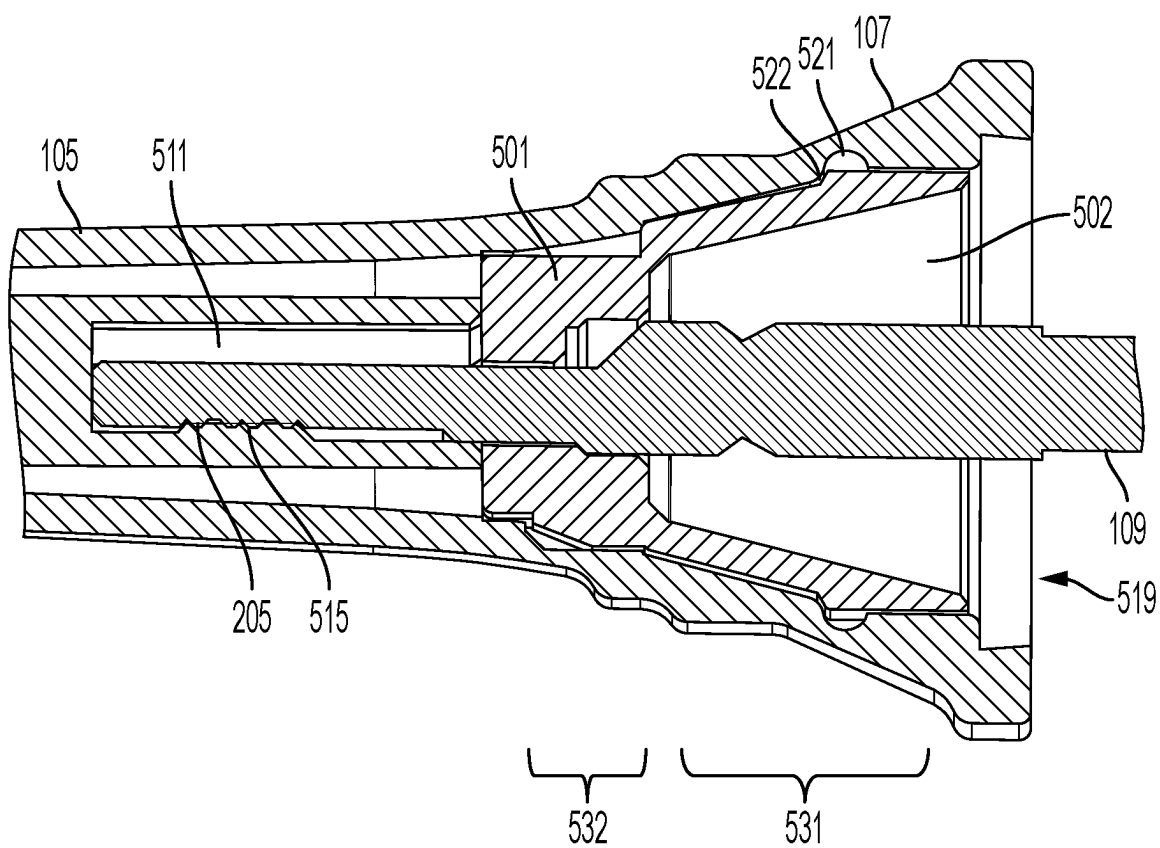


FIG. 5

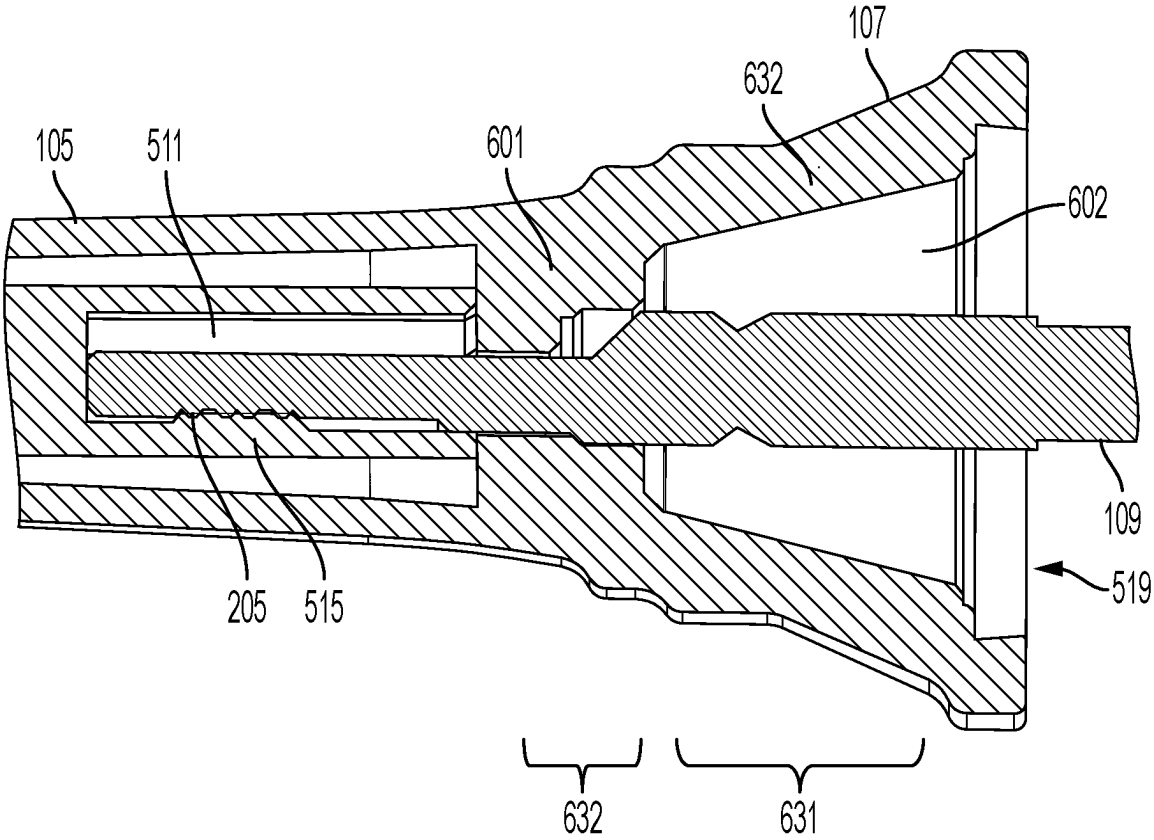


FIG. 6

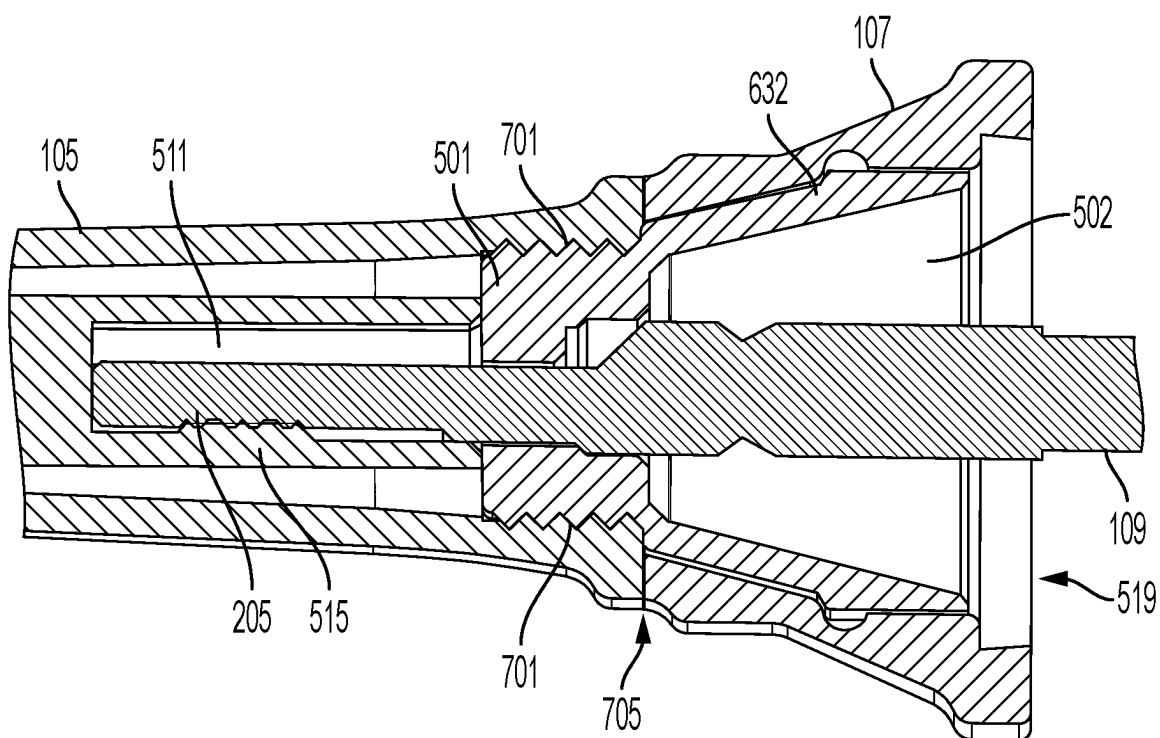


FIG. 7A



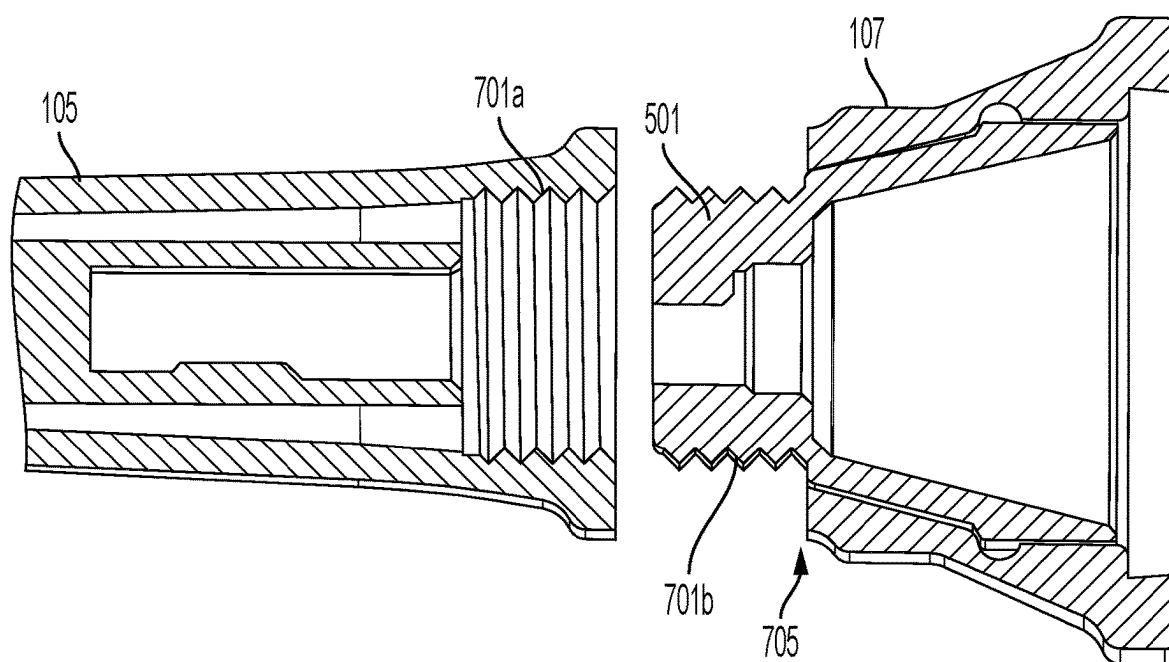


FIG. 7B

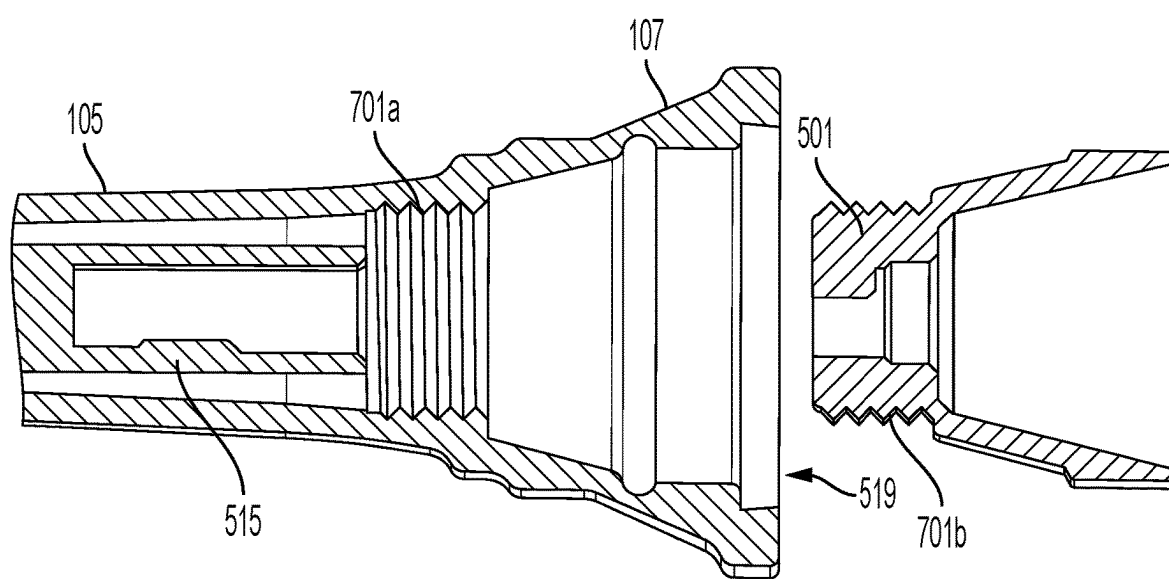


FIG. 8A

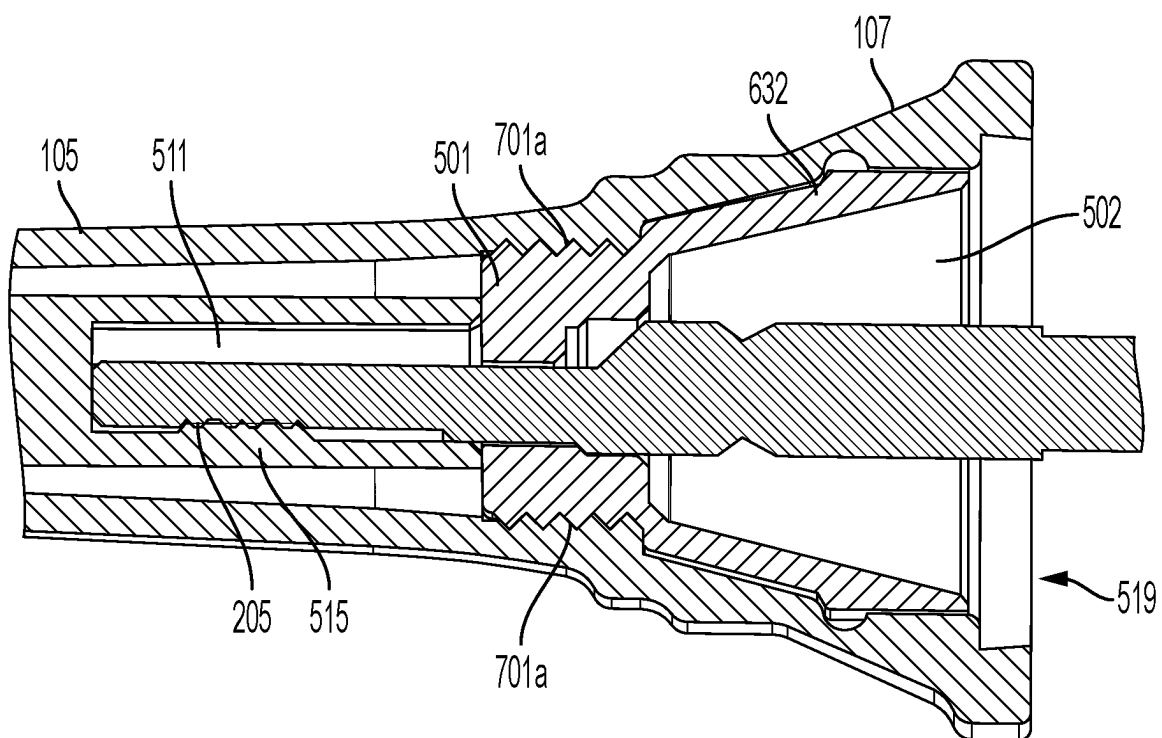


FIG. 8B

## REMOVABLE TOOTHBRUSH HEAD FOR ELECTRIC TOOTHBRUSH HANDLE

### BACKGROUND

[0001] This disclosure relates to the field of electric toothbrushes, and particularly brush heads that are configured to couple to an electric toothbrush handle.

[0002] Many types of electric toothbrushes use high speed vibrations of the motor inside the toothbrush handle to drive the brush head. In order to operate the electric toothbrush, the construction of the brush head requires that it be tightly coupled to the drive shaft of the toothbrush handle so it can withstand the high speed vibrations from operating the toothbrush. Therefore, many existing products use metal or springs inside the brush head to facilitate a tight coupling to the drive shaft of the toothbrush handle. This can increase the cost of making the brush head.

[0003] In addition, while some brush heads avoid the use of metal springs, those brush heads can sometimes have difficulty staying connected to the handle, and/or they may not offer as much vibratory effect due to a loose fit.

[0004] This patent document describes an apparatus that addresses at least some of the issues described above and/or other issues.

### SUMMARY

[0005] In a first aspect, a toothbrush head for an electric toothbrush includes a bristle tray having a plurality of bristles, a base having an opening for receiving a drive shaft of the electric toothbrush, a stem that connects the bristle tray to the base and that has an opening, and a coupling member. The coupling member includes an interior sidewall that surrounds a chamber for receiving the drive shaft, as well as a receptacle positioned to align with the opening of the stem. The coupling member does not include or rely on a spring to contact the drive shaft or the stem. A pin is positioned within the opening of the stem and the receptacle of the coupling member.

[0006] In a second aspect, a toothbrush head for an electric toothbrush includes: (a) a bristle tray that includes a plurality of bristles; (b) a base having an opening for receiving a drive shaft of an electric toothbrush; (c) a stem that connects the bristle tray to the base and that has an opening; and (d) a coupling member. The coupling member includes (a) a relatively narrower section positioned to align with the stem and (b) a relatively wider section positioned to align with the base. The relatively narrower section has a length that is shorter than that of the drive shaft so that, when the drive shaft is inserted into the coupling member, the drive shaft will extend from the relatively narrower section and into an interior cavity within the stem.

[0007] In a third aspect, a toothbrush comprises a handle with a drive shaft extending therefrom, and a toothbrush head. The toothbrush head includes a bristle tray comprising plurality of bristles, a base having an opening for receiving the drive shaft, a stem that connects the bristle tray to the base and that has an opening, and a coupling member. The coupling member includes a relatively narrower section positioned to align with the stem and a relatively wider section positioned to align with the base. The relatively narrower section has a length that is shorter than that of the drive shaft so that, when the drive shaft is inserted into the coupling member, the drive shaft will extend from the

relatively narrower section and into an interior cavity within the stem. The interior sidewall of the stem comprises an irregular surface that is positioned to contact the drive shaft when inserted into the toothbrush head. The drive shaft also comprises an irregular surface that is positioned to contact the interior sidewall of the stem when the drive shaft is inserted into the toothbrush head.

### BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of an electric toothbrush handle and removable toothbrush head.

[0009] FIGS. 2 and 3 illustrate a first embodiment of a toothbrush head with a coupling member according to the present disclosure.

[0010] FIG. 4 illustrates a variation of the embodiment of FIGS. 2 and 3.

[0011] FIG. 5 illustrates a second embodiment of a toothbrush head with a coupling member according to the present disclosure.

[0012] FIG. 6 illustrates a first variation of the embodiment of FIG. 5.

[0013] FIGS. 7A and 7B illustrate a second variation of the embodiment of FIG. 5.

[0014] FIGS. 8A and 8B illustrate a third variation of the embodiment of FIG. 5.

### DETAILED DESCRIPTION

[0015] As used in this document, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art. As used in this document, the term “comprising” (or “comprises”) means “including (or includes), but not limited to.” When used in this document, the term “exemplary” is intended to mean “by way of example” and is not intended to indicate that a particular exemplary item is preferred or required.

[0016] In this document, when terms such “first” and “second” are used to modify a noun, such use is simply intended to distinguish one item from another, and the terms not intended to require a sequential order unless specifically stated. The term “approximately,” when used in connection with a numeric value, is intended to include values that are close to, but not exactly, the number. For example, in some embodiments, the term “approximately” may include values that are within  $\pm 10$  percent of the value.

[0017] In this document, the term “connected”, when referring to two physical structures, means that the two physical structures touch each other. Devices that are connected may be secured to each other, or they may simply touch each other and not be secured.

[0018] When used in this document, terms such as “top” and “bottom,” “upper” and “lower”, or “front” and “rear,” are not intended to have absolute orientations but are instead intended to describe relative positions of various components with respect to each other. For example, a first component may be an “upper” component and a second component may be a “lower” component when a device of which the components are a part is oriented in a first direction. The relative orientations of the components may be reversed, or the components may be on the same plane, if the orientation of the structure that contains the components is changed.

The claims are intended to include all orientations of a device containing such components.

[0019] This disclosure is not limited to the particular systems, methodologies or protocols described, as these may vary. The terminology used in this description is for the purpose of describing the particular versions or embodiments only, and it is not intended to limit the scope.

[0020] In various embodiments, a brush head **101** is couplable to a drive shaft **109** of an electric toothbrush handle **102**, as shown in FIG. 1. The brush head **101** includes a bristle tray **106** that is a substrate from which multiple attached bristles extend. The brush head **101** also has a stem **105**, which extends from the bristle tray **106**, leading to a base **107** that flares outward from the stem. In use, the base will be placed over and attach to the drive shaft **109** of the electric toothbrush handle **102**. The handle **102** includes a motor and/or other electrical components that, when actuated, will cause the drive shaft **109** to vibrate. When the brush head **101** is attached to the drive shaft **109**, the vibration is transferred along the stem **105** to the bristle tray **106**, to facilitate brushing of the user's teeth.

[0021] The stem **105** and base **107** each have an opening, along with one or more internal structures, that are configured to receive and attach to the drive shaft **109**. The internal structures may be integral with the stem **105** and/or base **107**, such as part of a sidewall of an interior cavity of the stem **105** and/or base **107**. Alternatively or in addition, some or all of the internal structures may be formed in a coupling member that fits within the interior cavity. Examples of such structures will be described below.

[0022] One configuration is shown in FIGS. 2 and 3. FIG. 2 is a cross-sectional view, while FIG. 3 is an exterior view. In this embodiment, a coupling member **201** fits within the interior cavity **211** of the stem **105** and base **107**. The coupling member **201** includes a chamber **202** that includes a relatively wider section **231** that is positioned to align with the base **107** and a relatively narrower section **232** that is positioned to align with the stem **105**. Optionally, the interior sidewall of the relatively narrower section **232** may be cylindrical. Optionally, the interior sidewall of the relatively wider section **231** may be funnel-shaped with an opening aligned with the opening **219** and a sidewall having a taper that narrows as it gets closer to the interior sidewall of the relatively narrower section **232**.

[0023] When the drive shaft **109** is placed through the opening **219** of the brush head, it will be received into the chamber **202** and connect to a sidewall of the relatively narrower section **232** of the chamber **202**. Optionally, as shown the drive shaft may not contact any portion of the sidewall of the relatively wider section **231** of the chamber **202**.

[0024] The connection of the drive shaft **109** to the sidewall of the relatively narrower section **232** may be at least in part a friction fit. In some embodiments, the drive shaft **109** may include an irregular (i.e., textured and not smooth) surface **205** such as teeth, a gear, or a ledge that promotes a fit with the sidewall of chamber **202**. In some embodiments, the interior sidewall of the relatively narrower section **232** of the chamber **202** may include an irregular surface **215** such as a tooth or multiple teeth, a gear, or a ledge that promotes contact with the drive shaft **109**. In some embodiments, both the interior sidewall of the relatively narrower section **232** of the chamber **202** and the drive shaft **109** may have irregular surfaces **215**, **205** that are positioned to align with each

other. For example, each irregular surface **215**, **205** may include a gear in which the teeth of one irregular surface (such as surface **215**) are positioned to align with gaps between teeth of the other irregular surface (such as surface **205**). With such a connection, the coupling member does not include or rely on a spring to contact the drive shaft or the stem to transfer vibration from the drive shaft or provide support to the brush head on the drive shaft.

[0025] The height of the irregular surface of either element may be sufficient to provide a friction fit between the drive shaft and the interior sidewall of the relatively narrower section **232**, while still being separable with a moderate pulling force by a human. In this way, the brush head is removably attached to the drive shaft of the handle.

[0026] An end of the portion of the coupling member **201** that includes the relatively narrower section **232** may include a receptacle **204** that is positioned to align with an opening **203** of the stem **105**. The receptacle **204** may have a longitudinal axis that is substantially perpendicular to the longitudinal axis of the stem **107**. A pin **212** may be positioned within the opening **203** and the receptacle **204** to provide further stability and hold the coupling member in place. The pin may be made of metal such as aluminum, or it may be made of a rigid or semi-rigid plastic material.

[0027] FIG. 3 also shows the substrate **216** and bristles **217** of the bristle tray **106**.

[0028] In the embodiment of FIG. 2, the interior sidewall of the relatively narrower section **232** that surrounds the chamber **202** fully surrounds the chamber **202** without a break in the interior sidewall. FIG. 4 shows an alternate embodiment in which the interior sidewall includes a break **244** (i.e., an opening) so that part of the chamber **202** is open and faces the stem. Optionally, the break **244** may be positioned in other locations on the interior sidewall, and in some embodiments the interior sidewall may have multiple breaks.

[0029] FIG. 5 is a cross-sectional view of a second embodiment in which the coupling member **501** also fits within the interior cavity **511** of the stem **105** and base **107**. The coupling member **501** includes a chamber **502** that includes a relatively wider section **531** that is positioned to align with the base **107** and a relatively narrower section **532** that is positioned to align with the stem **105**. However, unlike the embodiment of FIGS. 2 and 3, the relatively narrower section **532** of the embodiment of FIG. 5 has a length that is shorter than that of the drive shaft **109**, and no pin secures the coupling member **501** in place. The drive shaft **109** therefore extends out from a top end of the relatively narrower section **532** and into the interior cavity **512** of the stem **105**.

[0030] The interior cavity **511** of the stem **105** may be cylindrical. When the drive shaft **109** is placed through the opening **519** of the brush head, it will be received into the interior cavity **511** of the stem **105** and connect to a sidewall of the interior cavity **511** of the stem **105**. Optionally, as shown the drive shaft may not contact any portion of the sidewall of the relatively wider section **531** of the chamber **502**.

[0031] The connection of the drive shaft **109** to the interior cavity **511** of the stem **105** may be at least in part a friction fit. In some embodiments, the drive shaft **109** may include an irregular (i.e., textured and not smooth) surface **205** such as teeth, a gear, or a ledge that promotes a fit with the sidewall of interior cavity **511**. In some embodiments, the

interior cavity **511** may include an irregular surface **515** such as a tooth or multiple teeth, a gear, or a ledge extends from the sidewall of the interior cavity **511** and promotes contact with the drive shaft **109**. In some embodiments, both the interior sidewall of the relatively narrower section **532** of the chamber **551** and the drive shaft **109** may have irregular surfaces **515**, **205** that are positioned to align with each other. For example, each irregular surface **515**, **205** may include a gear in which the teeth of one irregular surface (such as surface **515**) are positioned to align with gaps between teeth of the other irregular surface (such as surface **205**).

**[0032]** The height of the irregular surface of either element may be sufficient to provide a friction fit between the drive shaft and the interior sidewall of the relatively narrower section **532**, while still being separable with a moderate pull by a human. In this way, the brush head is removably attached to the drive shaft of the handle.

**[0033]** In this embodiment, the interior sidewall of the relatively wider section **531** may include a cylindrical first subsection near the opening **519**, followed by a funnel-shaped second subsection having a taper that narrows as it gets closer to the interior sidewall of the relatively narrower section **232**. The transition from the cylindrical first section to the tapered second subsection may occur at a corner **522**, which may include a ledge or ridge that is positioned to align with a corresponding corner **521**, which may be a ledge or ridge of the interior sidewall of the base **107** of the stem. The corner **521** of the interior sidewall of the base **107** of the stem thus serves as a stop to limit a path of travel of the coupling member into the interior cavity **511** of the stem **105**.

**[0034]** FIG. 6 illustrates a variation of FIG. 5 in which the coupling member **601** is integral with the stem **105** and base **107** and thus not a separate part. In FIG. 6, coupling member **601** narrows the portion of the interior cavity **511** that is positioned between the stem **105** and base **107**. The portion of the base **107** that includes the coupling member **601** includes a chamber **602** with includes a relatively wider section **631** that is positioned to align with the base **107**, and the portion of the stem **105** that includes the coupling member **601** includes a relatively narrower section **632** that is positioned to align with the stem **105**. All other elements of the embodiment of FIG. 6 are the same as those discussed above for FIG. 5.

**[0035]** FIGS. 7A and 7B illustrate a different variation of the embodiment of FIG. 5 in which the upper (relatively narrower) portion of coupling member **501** is coupled to the relatively narrower portion of interior of stem **105** by a set of threads **701**. FIG. 7A shows coupling member **501** attached to stem **105** by threads **701**, which forms a seam **705** between stem **105** and base **107**. FIG. 7b shows stem **105** and base **107** separated from each other, and that the interior sidewall of stem **105** includes threads **701a** that are sized and arranged to mate with threads **701b** that are positioned on the exterior sidewall of the upper portion of coupling member **501**.

**[0036]** FIGS. 8A and 8B illustrate a different variation of the embodiment of FIG. 5 in which, like embodiment of FIGS. 7A and 7B, the upper (relatively narrower) portion of coupling member **501** is coupled to the relatively narrower portion of interior of stem **105** by a set of threads **701**. However, in this embodiment stem **105** and base **107** do not separate from each other. Instead, as shown in FIG. 8A,

before assembly coupling member **501** begins outside of interior cavity **511**. In FIG. 8B, coupling member **501** is placed in cavity **511** and turned so that the threads **701b** on the exterior of the coupling member mate with threads **701a** that are positioned on the interior sidewall of the stem **105**.

**[0037]** Optionally, in the embodiments of FIGS. 7A-7B and 8A-8B, instead of positioning the threads on an interior sidewall of the stem, the threads could be positioned on an interior sidewall of the base. If so, then the threads on the coupling member may be positioned further down the sidewall than the figures here depict, so that the coupling member's threads are on the relatively wider portion of the coupling member. If this variation is used in the embodiment of FIGS. 7A and 7B, then the seam would not be positioned between the stem and the base but instead would be positioned somewhere on the base.

**[0038]** Notably, in each of the embodiments described above, no metal spring is used to transfer vibration from the drive shaft to the bristle tray. The embodiments described above eliminate the need for a metal spring. However, a spring may optionally be included in any of the embodiments.

**[0039]** In the various embodiments above, the coupling member and/or stem may be made of a resilient material, to provide a spring function that allows the drive shaft to vibrate when the coupling member or stem is positioned against the drive shaft. For example, at least the interior sidewall of the coupling member and/or stem may be formed of a soft plastic such as a low density polyethylene (LDPE), rubber, or other soft yet strong material. The entire coupling member or stem may be made of such material, or the gear or the interior sidewall of the stem or coupling member may be made of such materials.

**[0040]** Various aspects of the configurations described above may be combined with each other. For example. Other combinations are possible.

**[0041]** The disclosures of the following United States Patents and Patent Applications are incorporated into this document by reference: U.S. Pat. Nos. 9,724,180; 9,827,079; 11,890,153; US 2022/0273407; and US 2023/0338126.

**[0042]** In various embodiments, the exterior sidewalls of the coupling member may be smooth as shown in FIGS. 2-4. Alternatively, the exterior sidewalls of the coupling member may include ridges that are positioned to align with receptacles of the interior sidewall of the stem, or receptacles that are positioned to align with ridges of the interior sidewall of the stem, such as is disclosed in the patents incorporated by reference above.

**[0043]** The various embodiments disclosed in this patent document provide advantages over the prior art, whether standalone or combined. For example, the resilient wall of the handle coupling section of the coupling member that is positioned to engage with the drive shaft of the electric toothbrush handle may be made from the same wall of the handle coupling section of the coupling member, which requires no metal spring or other separate resilient member. This both achieves a tight coupling (because there is no separate part) and reduces the cost of making. Further, multiple engagements of various parts of the stem and the coupling member all facilitate a tight coupling between the coupling member and the stem, and thus improve the stability of the brush head as it is operating under the high speed vibrations of the motor inside the handle.

[0044] Further, the toothbrush head may be easy to manufacture as all the parts can be made of plastic or polyester materials. The cone shaped opening of the stem may facilitate easy receiving and coupling of the coupling member into the stem, and multiple engagement means, such as, the longitudinal ridge of the coupling member and the corresponding longitudinal channel in the stem, the mounting channel of the coupling member and the corresponding ridge block in the stem, the raised resilient wall of the coupling member and the corresponding coupling channel in the stem, the mounting block of the coupling member and the corresponding mounting channel in the stem, the base ring of the coupling member and the mating sitting platform in the stem, the mounting holes of the coupling member and the mounting ridges in the stem, each may facilitate easy sliding and positioning of the coupling member into the stem and secure mounting of the coupling member to the stem.

[0045] Other advantages of the present invention can be apparent to those skilled in the art from the foregoing specification. Accordingly, it will be recognized by those skilled in the art that changes or modifications may be made to the above-described embodiments without departing from the broad inventive concepts of the invention. It should therefore be understood that this invention is not limited to the particular embodiments described in this document, but is intended to include all changes and modifications that are within the scope and spirit of the invention as defined in the claims.

[0046] Various aspects of the invention are also described in the following clauses. The features of these clauses may be variously combined with features of others of these clauses, as well as with other features described above in this document.

[0047] Clause 1: A toothbrush head for an electric toothbrush includes a bristle tray having a plurality of bristles, a base having an opening for receiving a drive shaft of the electric toothbrush, a stem that connects the bristle tray to the base and that has an opening, and a coupling member. The coupling member includes an interior sidewall that surrounds a chamber for receiving the drive shaft, as well as a receptacle positioned to align with the opening of the stem. The coupling member does not include or rely on a spring to contact the drive shaft or the stem. A pin is positioned within the opening of the stem and the receptacle of the coupling member.

[0048] Clause 2: The toothbrush head of clause 1, wherein the coupling member comprises (a) a relatively narrower section positioned to align with the stem, and (b) a relatively wider section positioned to align with the base.

[0049] Clause 3: The toothbrush head of clause 2, wherein an interior sidewall of the relatively narrower section comprises an irregular surface that is positioned to contact the drive shaft when inserted into the toothbrush head.

[0050] Clause 4: The toothbrush head of clause 2 or 3, further comprising the electric toothbrush, and wherein the drive shaft comprises an irregular surface that is positioned to contact an interior sidewall of the relatively narrower section when the drive shaft is inserted into the toothbrush head.

[0051] Clause 5: The toothbrush head of any of clauses 2-4, further comprising the electric toothbrush, and

wherein: (a) the drive shaft comprises a first gear that is positioned to contact an interior sidewall of the relatively narrower section when the drive shaft is inserted into the toothbrush head; (b) the interior sidewall of the relatively narrower section comprises a second gear comprising teeth that extend toward the drive shaft; and (c) the teeth of the second gear are positioned to align with gaps between teeth of the first gear.

[0052] Clause 6: The toothbrush head of clause 2 or 3, further comprising the electric toothbrush, and wherein: (a) the drive shaft comprises a gear that is positioned to contact an interior sidewall of the relatively narrower section when the drive shaft is inserted into the toothbrush head; (b) the interior sidewall of the relatively narrower section comprises a ledge that extends toward the drive shaft; and (c) the ledge is positioned to align with the gear.

[0053] Clause 7: The toothbrush head of any of clauses 1-5, wherein the interior sidewall that surrounds the cavity fully surrounds the chamber without a break in the interior sidewall.

[0054] Clause 8: A toothbrush head for an electric toothbrush includes: (a) a bristle tray that includes a plurality of bristles; (b) a base having an opening for receiving a drive shaft of an electric toothbrush; (c) a stem that connects the bristle tray to the base and that has an opening; and (d) a coupling member. The coupling member includes (a) a relatively narrower section positioned to align with the stem and (b) a relatively wider section positioned to align with the base. The relatively narrower section has a length that is shorter than that of the drive shaft so that, when the drive shaft is inserted into the coupling member, the drive shaft will extend from the relatively narrower section and into an interior cavity within the stem.

[0055] Clause 9: The toothbrush head of clause 8, wherein the interior sidewall of the stem comprises an irregular surface that is positioned to contact the drive shaft when inserted into the toothbrush head.

[0056] Clause 10: The toothbrush head of clause 8 or 9, further comprising the electric toothbrush, and wherein the drive shaft comprises an irregular surface that is positioned to contact an interior sidewall of the stem when the drive shaft is inserted into the toothbrush head.

[0057] Clause 11: The toothbrush head of clause 8 or 9, further comprising the electric toothbrush, and wherein: (a) the drive shaft comprises a first gear that is positioned to contact an interior sidewall of the stem when the drive shaft is inserted into the toothbrush head; (b) the interior sidewall of the stem comprises a second gear comprising teeth that extend toward the drive shaft; and (c) the teeth of the second gear are positioned to align with gaps between teeth of the first gear.

[0058] Clause 12: The toothbrush head of any of clauses 8-11, wherein: (a) the relatively wider section includes a cylindrical first subsection and a funnel-shaped second subsection having a taper that narrows as it gets closer to the relatively narrower section; (b) an exterior sidewall of the relatively wider section includes a corner positioned at a point at which the first subsection transitions to the second subsection; and (c) the base

includes a corresponding corner that is positioned to limit a path of travel of the coupling member into the interior cavity of the stem.

[0059] Clause 13: The toothbrush head of any of clauses 8-12, wherein the coupling member is integrally formed with the base.

[0060] Clause 14: The toothbrush head of any of clauses 8-13, wherein: (a) an exterior sidewall of the coupling member includes threads; (b) an interior sidewall of the stem or the base includes threads; and (c) the threads of the coupling member are configured to mate with the threads of the stem or the base when the coupling member is threaded into the stem or the base.

[0061] Clause 15: The toothbrush head of clause 14, further including a seam on the stem, on the base, or between the stem and the base, wherein the seam is positioned to align with the threads of the stem or the base.

[0062] Clause 16: A toothbrush comprises a handle with a drive shaft extending therefrom, and a toothbrush head. The toothbrush head includes a bristle tray comprising plurality of bristles, a base having an opening for receiving the drive shaft, a stem that connects the bristle tray to the base and that has an opening, and a coupling member. The coupling member includes a relatively narrower section positioned to align with the stem and a relatively wider section positioned to align with the base. The relatively narrower section has a length that is shorter than that of the drive shaft so that, when the drive shaft is inserted into the coupling member, the drive shaft will extend from the relatively narrower section and into an interior cavity within the stem. An interior sidewall of the stem comprises an irregular surface that is positioned to contact the drive shaft when inserted into the toothbrush head. The drive shaft also comprises an irregular surface that is positioned to contact the interior sidewall of the stem when the drive shaft is inserted into the toothbrush head.

[0063] Clause 17: The toothbrush of clause 16 wherein: (a) the irregular surface of the drive shaft comprises a first gear that is positioned to contact the interior sidewall of the stem when the drive shaft is inserted into the toothbrush head; (b) the irregular surface of the stem comprises a second gear comprising teeth that extend toward the drive shaft; and (c) the teeth of the second gear are positioned to align with gaps between teeth of the first gear.

[0064] Clause 18: The toothbrush of clause 16 or 17, wherein: (a) the relatively wider section includes a cylindrical first subsection and a funnel-shaped second subsection having a taper that narrows as it gets closer to the relatively narrower section; (b) an exterior sidewall of the relatively wider section comprises a corner positioned at a point at which the cylindrical first section transitions to the tapered second subsection; and (c) the base includes a corresponding corner that is positioned to limit a path of travel of the coupling member into the interior cavity of the stem.

[0065] Clause 19: The toothbrush of any of clauses 16-18, wherein the coupling member is integrally formed with the base.

[0066] Clause 20: The toothbrush of any of clauses 16-19, further comprising a seam on the stem, on the

base, or between the stem and the base, wherein the seam is positioned to align with threads of the stem or the base.

1. A toothbrush head for an electric toothbrush, the toothbrush head comprising:

- a bristle tray comprising a plurality of bristles;
- a base having an opening for receiving a drive shaft of an electric toothbrush;
- a stem that connects the bristle tray to the base and that has an opening;
- a coupling member, wherein the coupling member:
  - comprises an interior sidewall that surrounds a chamber for receiving the drive shaft,
  - includes a receptacle positioned to align with the opening of the stem, and
  - does not include or rely on a spring to contact the drive shaft or the stem; and
- a pin positioned within the opening of the stem and the receptacle of the coupling member.

2. The toothbrush head of claim 1, wherein the coupling member comprises:

- a relatively narrower section positioned to align with the stem; and
- a relatively wider section positioned to align with the base.

3. The toothbrush head of claim 2, wherein an interior sidewall of the relatively narrower section comprises an irregular surface that is positioned to contact the drive shaft when inserted into the toothbrush head.

4. The toothbrush head of claim 2 further comprising the electric toothbrush, and wherein the drive shaft comprises an irregular surface that is positioned to contact an interior sidewall of the relatively narrower section when the drive shaft is inserted into the toothbrush head.

5. The toothbrush head of claim 2 further comprising the electric toothbrush, and wherein:

- the drive shaft comprises a first gear that is positioned to contact an interior sidewall of the relatively narrower section when the drive shaft is inserted into the toothbrush head;
- the interior sidewall of the relatively narrower section comprises a second gear comprising teeth that extend toward the drive shaft; and
- the teeth of the second gear are positioned to align with gaps between teeth of the first gear.

6. The toothbrush head of claim 2 further comprising the electric toothbrush, and wherein:

- the drive shaft comprises a gear that is positioned to contact an interior sidewall of the relatively narrower section when the drive shaft is inserted into the toothbrush head;
- the interior sidewall of the relatively narrower section comprises a ledge that extends toward the drive shaft; and
- the ledge is positioned to align with the gear.

7. The toothbrush head of claim 1, wherein the interior sidewall that surrounds the cavity fully surrounds the chamber without a break in the interior sidewall.

8. A toothbrush head for an electric toothbrush, the toothbrush head comprising:

- a bristle tray comprising a plurality of bristles;
- a base having an opening for receiving a drive shaft of an electric toothbrush;



a stem that connects the bristle tray to the base and that has an opening;

a coupling member, wherein:

- the coupling member includes a relatively narrower section positioned to align with the stem and a relatively wider section positioned to align with the base, and
- the relatively narrower section has a length that is shorter than that of the drive shaft so that, when the drive shaft is inserted into the coupling member, the drive shaft will extend from the relatively narrower section and into an interior cavity within the stem.

9. The toothbrush head of claim 8, wherein the interior sidewall of the stem comprises an irregular surface that is positioned to contact the drive shaft when inserted into the toothbrush head.

10. The toothbrush head of claim 8 further comprising the electric toothbrush, and wherein the drive shaft comprises an irregular surface that is positioned to contact an interior sidewall of the stem when the drive shaft is inserted into the toothbrush head.

11. The toothbrush head of claim 8 further comprising the electric toothbrush, and wherein:

- the drive shaft comprises a first gear that is positioned to contact an interior sidewall of the stem when the drive shaft is inserted into the toothbrush head;
- the interior sidewall of the stem comprises a second gear comprising teeth that extend toward the drive shaft; and
- the teeth of the second gear are positioned to align with gaps between teeth of the first gear.

12. The toothbrush head of claim 8, wherein:

- the relatively wider section includes a cylindrical first subsection and a funnel-shaped second subsection having a taper that narrows as it gets closer to the relatively narrower section;
- an exterior sidewall of the relatively wider section includes a corner positioned at a point at which the first subsection transitions to the second subsection; and
- the base includes a corresponding corner that is positioned to limit a path of travel of the coupling member into the interior cavity of the stem.

13. The toothbrush head of claim 8, wherein the coupling member is integrally formed with the base.

14. The toothbrush head of claim 8, wherein:

- an exterior sidewall of the coupling member comprises threads;
- an interior sidewall of the stem or the base comprises threads; and
- the threads of the coupling member are configured to mate with the threads of the stem or the base when the coupling member is threaded into the stem or the base.

15. The toothbrush head of claim 14, further comprising a seam on the stem, on the base, or between the stem and the base, wherein the seam is positioned to align with the threads of the stem or the base.

16. A toothbrush, comprising:

- a handle with a drive shaft extending therefrom; and
- a toothbrush head comprising:
  - a bristle tray comprising plurality of bristles,
  - a base having an opening for receiving the drive shaft,
  - a stem that connects the bristle tray to the base and that has an opening,
  - a coupling member, wherein:
    - the coupling member includes a relatively narrower section positioned to align with the stem and a relatively wider section positioned to align with the base; and
    - the relatively narrower section has a length that is shorter than that of the drive shaft so that, when the drive shaft is inserted into the coupling member, the drive shaft will extend from the relatively narrower section and into an interior cavity within the stem,

wherein:

- an interior sidewall of the stem comprises an irregular surface that is positioned to contact the drive shaft when inserted into the toothbrush head, and
- the drive shaft comprises an irregular surface that is positioned to contact the interior sidewall of the stem when the drive shaft is inserted into the toothbrush head.

17. The toothbrush of claim 16 wherein:

- the irregular surface of the drive shaft comprises a first gear that is positioned to contact the interior sidewall of the stem when the drive shaft is inserted into the toothbrush head;
- the irregular surface of the stem comprises a second gear comprising teeth that extend toward the drive shaft; and
- the teeth of the second gear are positioned to align with gaps between teeth of the first gear.

18. The toothbrush of claim 16, wherein:

- the relatively wider section includes a cylindrical first subsection and a funnel-shaped second subsection having a taper that narrows as it gets closer to the relatively narrower section;
- an exterior sidewall of the relatively wider section comprises a corner positioned at a point at which the cylindrical first section transitions to the tapered second subsection; and
- the base includes a corresponding corner that is positioned to limit a path of travel of the coupling member into the interior cavity of the stem.

19. The toothbrush of claim 16, wherein the coupling member is integrally formed with the base.

20. The toothbrush of claim 16, further comprising a seam on the stem, on the base, or between the stem and the base, wherein the seam is positioned to align with threads of the stem or the base.

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